

Section 5.1: Simple and Compound Interest

Finite Mathematics MTH 132 Spring 2014

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Mathematics of Finance

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Chapter 5: *Mathematics of Finance*

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5.1 Simple and Compound Interest

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5.2 Future Value of an Annuity

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5.3 Present Value of an Annuity; Amortization

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Example 5.1

If you borrow \$7200,
at an annual interest
rate of 6.2%
compounded daily,
how much is owed
after 15 months?

Chapter 5: *Mathematics of Finance*

5.1 Simple and Compound Interest

5.2 Future Value of an Annuity

5.3 Present Value of an Annuity; Amortization

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If you borrow \$7200, at an annual interest rate of 6.2% compounded daily, how much is owed after 15 months?

Example 5.2

You need to save \$250,000 for retirement in 20 years. What size should the monthly payments into a fund which pays 7% compounded monthly be?

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5.2 Future Value of an Annuity

5.3 Present Value of an Annuity; Amortization

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Example 5.2

You need to save \$250,000 for retirement in 20 years. What size should the monthly payments into a fund which pays 7% compounded monthly be?

Example 5.3

You have \$8,500 in credit card debt at 25% compounded monthly. If you want to pay off the debt on 5 years, how large should the monthly payments be?

Simple Interest

Simple Interest

Example 5.1.1.

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Simple Interest

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John borrows \$2,000 at 6.3% for 39 weeks. How much interest did he accrue?

Simple Interest

$$I = Prt$$

P is the *principal*

r is the annual interest
rate

t is the *time* in years

Simple Interest

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Example 5.1.2.

John borrows \$2,000 at 6.3% for 39 weeks. How much does he owe?

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Future/Maturity Value for Simple Interest

$$A = P(1 + rt)$$

Simple Interest

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John borrows \$2,000 at 6.3% for 39 weeks. How much does he owe?

Future/Maturity Value for Simple Interest

$$A = P(1 + rt)$$

A is the future *amount*.

Simple Interest

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Example 5.1.3.

A CD pays 3.75% after 5 years. If John wants to get \$5,000 from the CD how much should he invest?

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Present Value for Simple Interest

$$P = \frac{A}{1+rt}$$

Compound Interest

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Example 5.1.4.

John borrows \$2,000 at 6.3%
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How much does he owe?

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Future/Maturity Value for Compound Interest

$$A = P(1 + i)^n$$

Compound Interest

Example 5.1.4.

John borrows \$2,000 at 6.3%
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How much does he owe?

Future/Maturity Value for Compound Interest

$$A = P(1 + i)^n$$

$$i = \frac{r}{m} \text{ and } n = mt$$

Compound Interest

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Compound Interest

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m is the number of
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year.

n is the number of
compounding periods.

i is the *interest rate per*
period.

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Example 5.1.5.

An account pays 3.75% compounded semi-annually. If John wants to get \$5,000 in 5 years from the account how much should he invest?

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Present Value for Compound Interest

$$P = \frac{A}{(1+i)^n}$$

$$i = \frac{r}{m} \text{ and } n = mt$$

m is the number of *compounding periods per year*.

n is the number of *compounding periods*.

i is the *interest rate per period*.

Determining m

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- Biannually/semi-annually $\Rightarrow m = 2$
- Quarterly $\Rightarrow m = 4$
- Monthly $\Rightarrow m = 12$
- Weekly $\Rightarrow m = 52$

Determining m

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- Biannually/semi-annually $\Rightarrow m = 2$
- Quarterly $\Rightarrow m = 4$
- Monthly $\Rightarrow m = 12$
- Weekly $\Rightarrow m = 52$
- Daily $\Rightarrow m = 365$ for *exact* or 360 for *ordinary interest*.

Determining m

- Yearly $\Rightarrow m = 1$
- Biannually/semi-annually $\Rightarrow m = 2$
- Quarterly $\Rightarrow m = 4$
- Monthly $\Rightarrow m = 12$
- Weekly $\Rightarrow m = 52$
- Daily $\Rightarrow m = 365$ for *exact* or 360 for *ordinary interest*.
- Continuously $\Rightarrow m = \infty$

Continuous Compounding

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Example 5.1.6.

John borrows \$2,000 at 6.3%
compounded continuously for 39
weeks. How much does he owe?

Continuous Compounding

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John borrows \$2,000 at 6.3% *compounded continuously* for 39 weeks. How much does he owe?

Future/Maturity Value for
Continuously Compound
Interest

$$A = Pe^{rt}$$

Continuous Compounding

Example 5.1.6.

John borrows \$2,000 at 6.3% *compounded continuously* for 39 weeks. How much does he owe?

The number e

$$e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n$$
$$\approx 2.718281$$

Future/Maturity Value for
Continuously Compound
Interest

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Continuous Compounding

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Example 5.1.7.

John owes \$4,000 at 5%
compounded continuously after 10
years. How much did he borrow?

Continuous Compounding

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Present Value for Continuously
Compound Interest

$$P = \frac{A}{e^{rt}} = Ae^{-rt}$$

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The *effective rate* corresponding to a stated rate of interest r compounded continuously is

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- In the “real world” effective rate is usually called *APR* or *APY*.